Clipcademy review study · Form B

Participant code: _____ Date: _____

10 questions. Circle one letter for each. Don't use notes or your phone. Your answers don't affect your grade.

1. A template strand is drawn with its 5′ end on the left and its 3′ end on the right. Which way does the new strand grow as it's copied?

(A) Toward the right end
(B) Outward from the middle
(C) Toward the left end
(D) It can't be copied

2. A new DNA strand reads 5′-ACG-3′. Then T is added, then A. What does it read now?

(A) 5′-ATACG-3′
(B) 5′-ACGTA-3′
(C) 5′-TAACG-3′
(D) 5′-ACGAT-3′

3. In a cell, primase stops working but DNA polymerase is normal. Why can't the cell copy its DNA?

(A) There's no primer to build onto
(B) The template strand can't be read
(C) New strands would grow 3′ to 5′
(D) DNA polymerase runs out of nucleotides

4. A cell's mismatch repair proteins don't work, but its DNA polymerase is normal. When can a wrong base still be fixed?

(A) Before copying starts
(B) Only after copying is finished
(C) Never, in this cell
(D) While the DNA is being copied

5. A gene has a UV bulge in its backbone that was never repaired. What happens when the cell tries to transcribe that gene?

(A) It's read from the other end
(B) It can't be read normally
(C) The bulge is copied into the RNA
(D) Only the coding strand is read

6. Which order of steps repairs UV damage?

(A) Cut out, recognize, replace, reseal
(B) Recognize, replace, cut out, reseal
(C) Recognize, cut out, replace, reseal
(D) Recognize, reseal, cut out, replace

7. An NTP carrying A enters RNA polymerase's active site. The next base on the template strand is G. What happens?

(A) It is joined to the growing RNA
(B) It isn't joined, because it doesn't match
(C) It pairs with the coding strand instead
(D) It waits for the coding strand to be read

8. RNA polymerase is present, but it never starts transcribing gene G. Which part of gene G is most likely damaged?

(A) Its start codon
(B) Its coding strand
(C) Its anticodon
(D) Its promoter

9. A tRNA's anticodon is AAG. Which mRNA codon does it pair with? (Bases are listed in the order they pair, like AUG and UAC in the notes.)

(A) AAG
(B) UUC
(C) GAA
(D) UUG

10. In what order does one tRNA pass through the ribosome's sites?

(A) A, then P, then E
(B) P, then A, then E
(C) E, then A, then P
(D) A, then E, then P

Moderator only: score _____ / 10